WORKSHEET 1.3

Limits from Numerical Evidence

Unit 1: Limits and Continuity

Key ideas

- Estimate $\lim_{x\to c} f(x)$ from a table by looking at inputs approaching c from **both** sides.
- Read the *trend* of the outputs, not the single entry nearest to c.
- A table is evidence, not proof: write "the table suggests…".
- For trigonometric functions the calculator must be in **radian** mode.

Worked examples

Example 1 · Estimating a limit

Selected values of f are shown. Estimate $\lim_{x\to 4} f(x)$.

x	3.9	3.99	4.01	4.1
$f(x)$	1.9	1.99	2.01	2.1

Solution. From the left the values rise toward 2; from the right they fall toward 2.

The table suggests the limit is 2.

Example 2 · Sides that disagree

Selected values of g are shown. What does the table suggest about $\lim_{x\to 1} g(x)$?

x	0.9	0.99	1.01	1.1
$g(x)$	-1.2	-1.02	3.03	3.3

Solution. Left-hand values approach -1; right-hand values approach 3.

The table suggests that the two-sided limit does not exist.

Example 3 · Building a table (calculator)

Estimate $\lim_{x\to 0} \dfrac{e^{3x}-1}{x}$ to three decimal places.

Solution. At $x = 0.001$ the value is 3.0045; at $x = -0.001$ it is 2.9955. At $x = \pm 0.0001$ the values are 3.00045 and 2.99955.

The limit is approximately 3.000.

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Practice

CORE

1. A table gives $f(5.9) = 11.8$, $f(5.99) = 11.98$, $f(6.01) = 12.02$, $f(6.1) = 12.2$. Estimate $\lim_{x \to 6} f(x)$.

2. A table gives $h(1.9) = 6.7$, $h(1.99) = 6.97$, $h(2) = -4$, $h(2.01) = 7.03$, $h(2.1) = 7.3$. Estimate $\lim_{x \to 2} h(x)$.

3. A table gives values of k only at $x = 2.9,\ 2.99,\ 2.999$, which approach 8. What can and cannot be concluded?

4. Which inputs would you choose to investigate $\lim_{x \to -3} f(x)$? Give six.

EXAM LEVEL

5. (Calculator) Estimate $\lim_{x \to 0} \dfrac{\sin(2x)}{\sin(5x)}$ to three decimal places.

6. (Calculator) Estimate $\lim_{x \to 1} \dfrac{x^5 - 1}{x - 1}$.

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7. A student finds $\dfrac{\sin x}{x} \approx 0.01745$ at $x = 0.001$. What has gone wrong, and what is the correct value?

8. A function f is defined for all real numbers. A table shows $f(2.9) = 5.8$, $f(2.99) = 5.98$, $f(3.01) = 6.02$ and $f(3.1) = 6.2$. Must $\lim\limits_{x \to 3} f(x) = 6$? Explain.

Question 9 · Free-response practice

(Calculator) Let $h(x) = \dfrac{\sqrt{x+1} - 1}{x}$ for $x \neq 0$.

(a) Find $h(-0.1)$, $h(-0.01)$, $h(0.01)$ and $h(0.1)$ to four decimal places.

(b) Use your values to estimate $\lim\limits_{x \to 0} h(x)$.

(c) Explain why part (b) is not a proof, and confirm the limit algebraically.

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Worked solutions

1. 12.
2. 7. The entry $h(2) = -4$ is the function value, not the limit.
3. The table suggests $\lim_{x \to 3^-} k(x) = 8$. It gives no information about the right-hand limit, so nothing can be said about the two-sided limit.
4. For example $-3.1,\ -3.01,\ -3.001,\ -2.999,\ -2.99,\ -2.9$: three on each side, closing in.
5. At $x = \pm 0.01$ the value is 0.40014; at ± 0.001 it is 0.4000014. The limit is about 0.400.
6. At $x = 1.001$: 5.0100. At $x = 0.999$: 4.9900. The limit is about 5.
7. The calculator is in degree mode ($0.01745 \approx \frac{\pi}{180}$). In radian mode the value is 0.9999998, close to 1.
8. No. Four values suggest the limit is 6 but cannot force it; f is unrestricted at inputs closer to 3 than those listed.
9. (a) $h(-0.1) = 0.5132$, $h(-0.01) = 0.5013$, $h(0.01) = 0.4988$, $h(0.1) = 0.4881$.
   (b) From both sides the values approach 0.5.
   (c) A table samples only finitely many inputs. Algebraically, $\dfrac{\sqrt{x+1}-1}{x} \cdot \dfrac{\sqrt{x+1}+1}{\sqrt{x+1}+1} = \dfrac{1}{\sqrt{x+1}+1} \to \dfrac{1}{2}$.

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